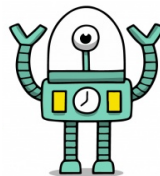
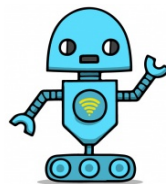


30/09/2026

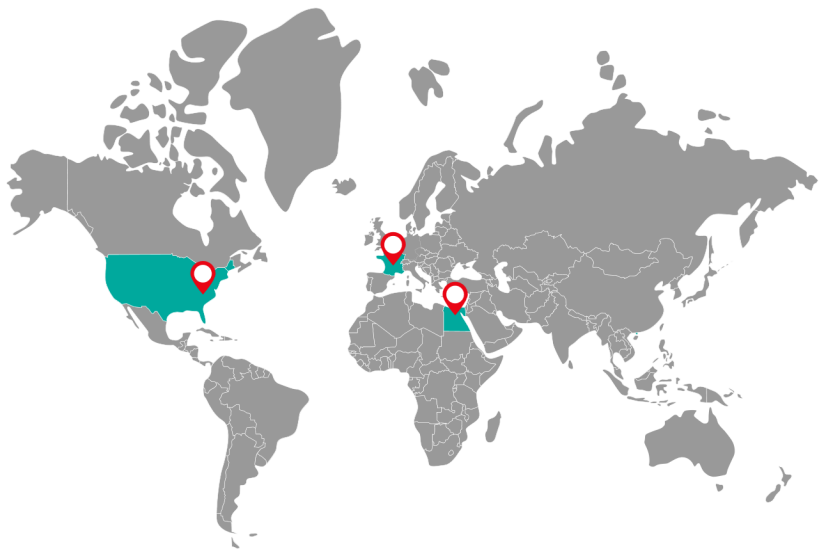
Pyladies

Intro to Reinforcement Learning

Ingy ElSayed-Aly, PhD

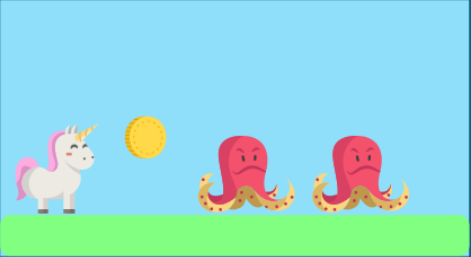
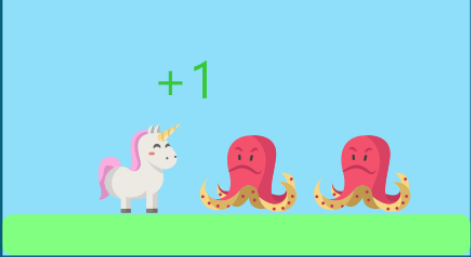


About me

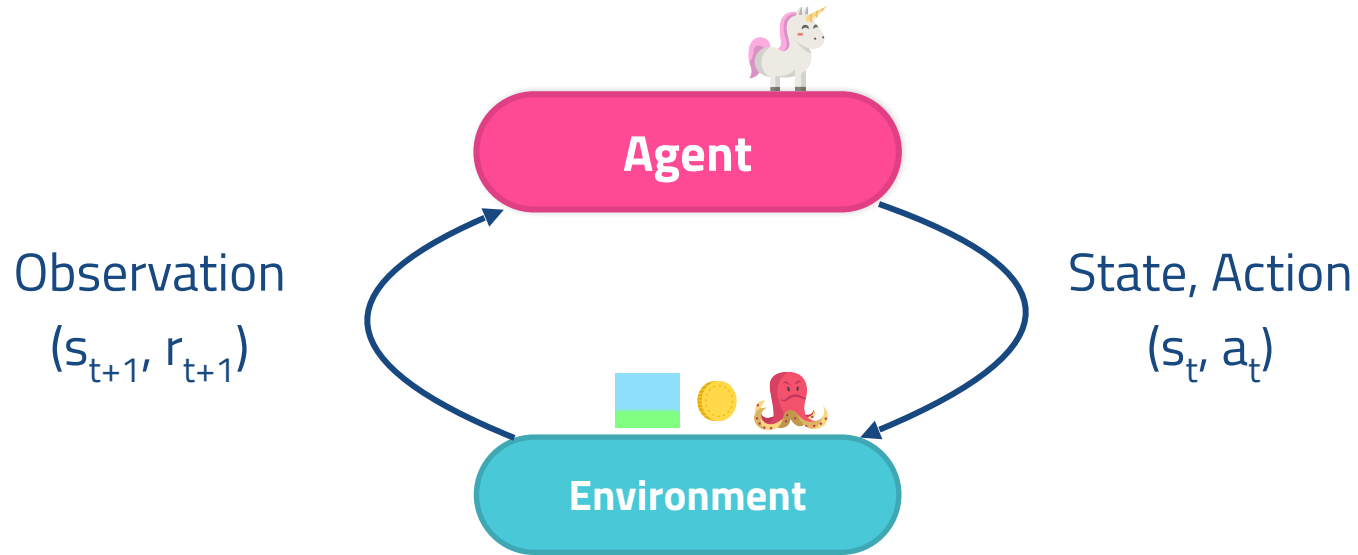
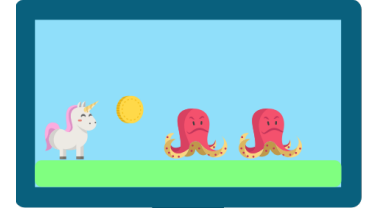


- Computer Science Bachelor at the American University in Cairo, Egypt
- Master & PhD in Computer Science at the University of Virginia, Charlottesville, USA
- Lead AI Engineer - REEV, Toulouse, France
- Ingénieure de Recherche - Saint-Gobain Recherche, Aubervilliers, France

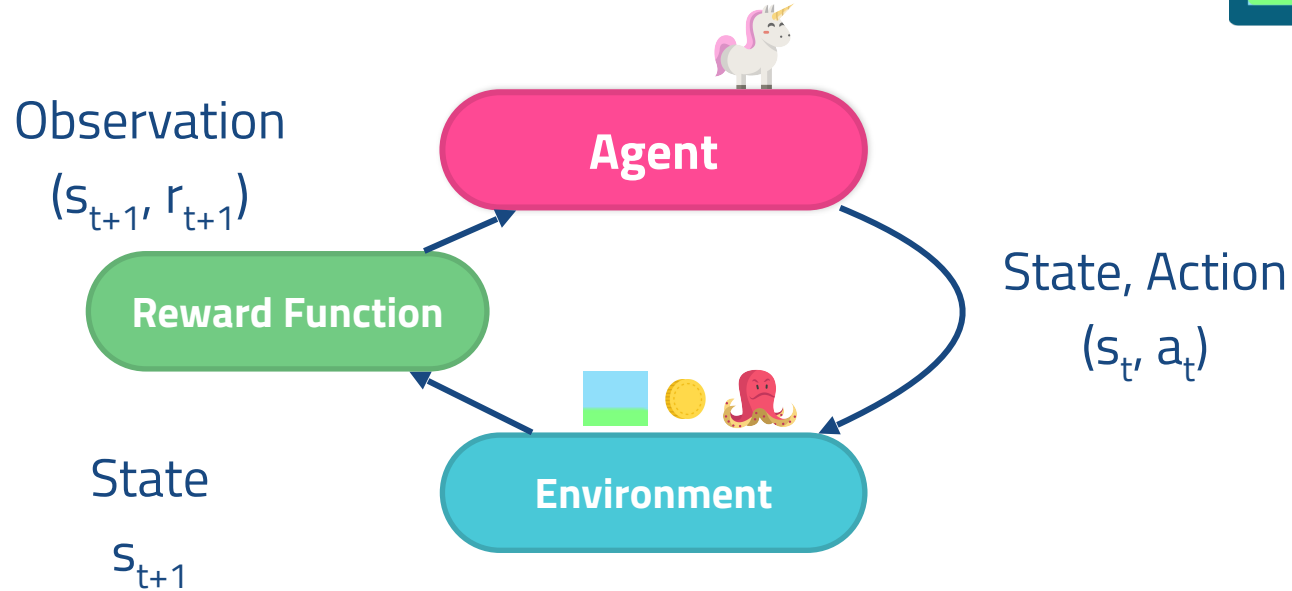
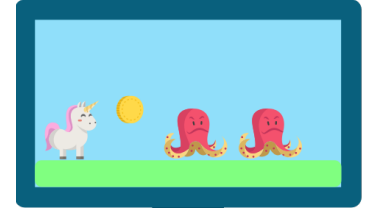
Reinforcement Learning

Available Actions	Available Actions	Available Actions
		

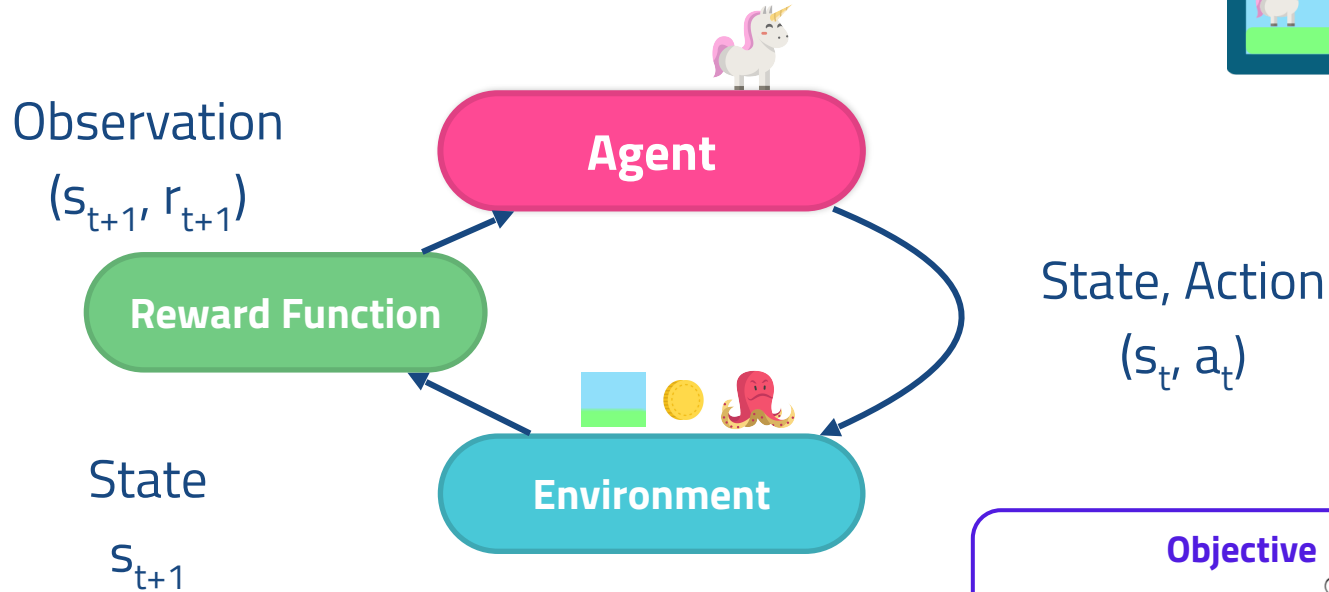
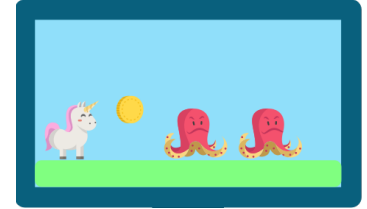
Reinforcement Learning



Reinforcement Learning



Reinforcement Learning

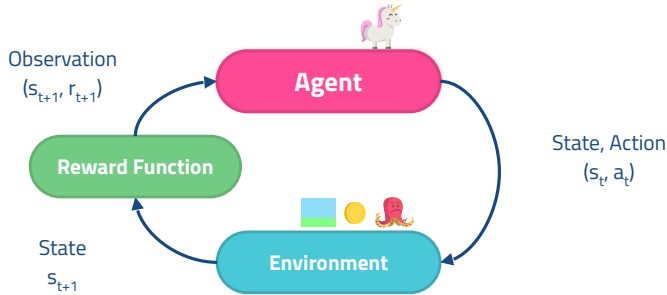
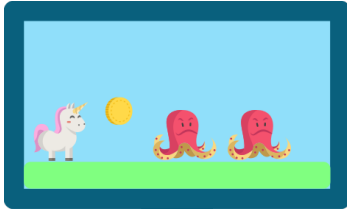


Objective

$$\max R(\tau) = \sum_{k=0}^{\infty} \gamma^k r_t$$

Reinforcement Learning

State Space, Action Space, Observations



- State Space : where everything is.

○     $S_0 = [(0,0), (1,0), (2,0), (3,0)] \rightarrow (x,y)$

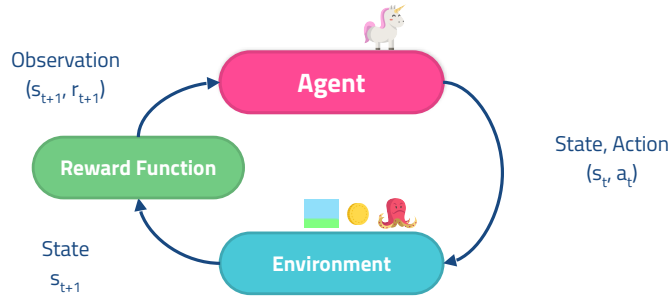
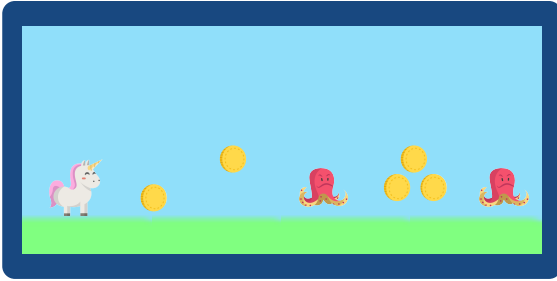
- Action Space : what actions can the agent take?

○  $A_0 = [\text{Jump, Right}]$

- Observation : what our agent “sees”, when $A_0 = \text{Right}$

○    $O_1 = [((1,0), ()), (2,0)), 1]$

Reinforcement Learning Rewards & Discounting



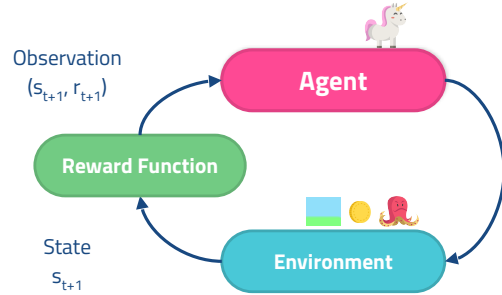
- Reward : when $A_0 = \text{Right}$, $R_1=1$, $R_3=-5$
- Cumulative Reward : Total reward during an episode (until the unicorn hits an enemy or runs out of time).
- Discounted Total Reward: Total reward multiplied by $\gamma \in [0,1]$. Controls how much value you give to rewards that take longer to get to.

Objective

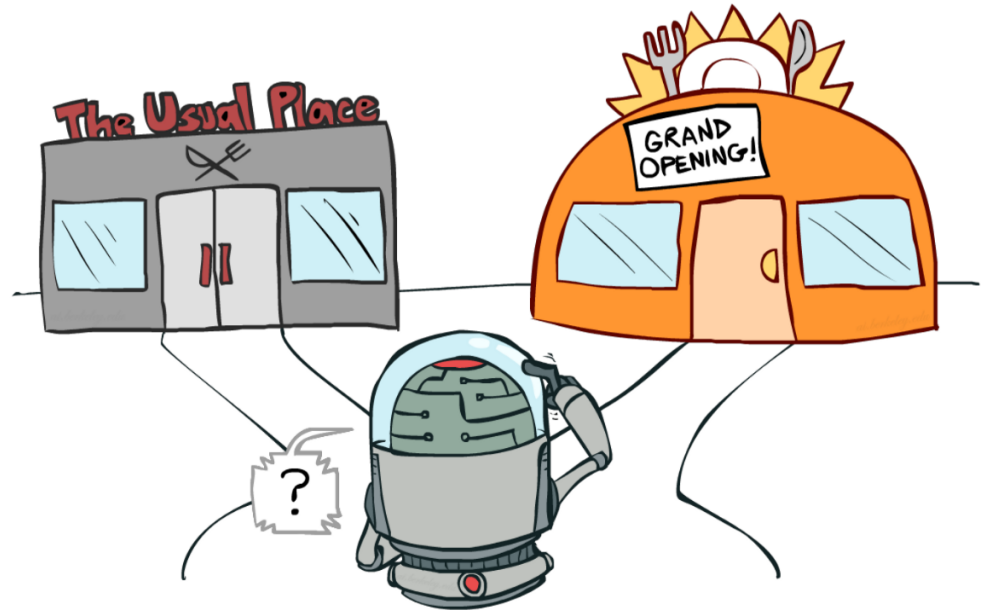
$$\max R(\tau) = \sum_{k=0}^{\infty} \gamma^k r_t$$

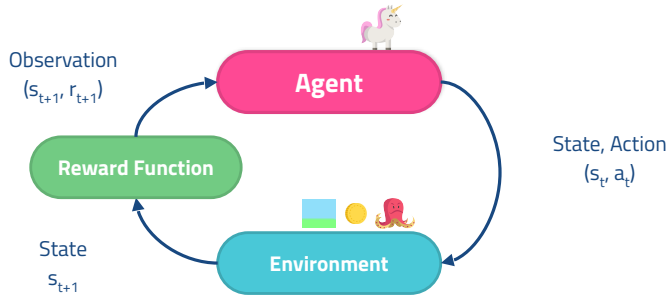
Reinforcement Learning

Exploration vs Exploitation



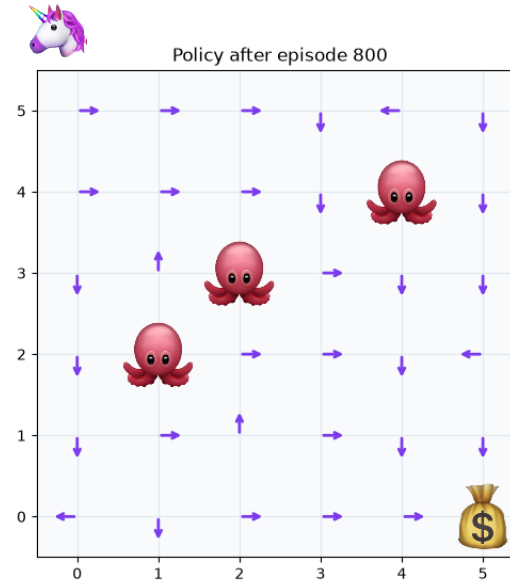
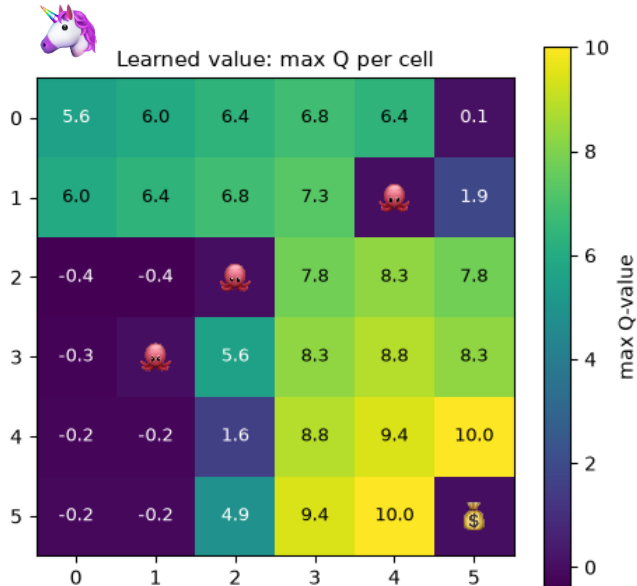
State, Action
 (s_t, a_t)





Reinforcement Learning Q-Learning

- Save a table for each state, action pair
- Estimate the Value of an action based on discounted rewards



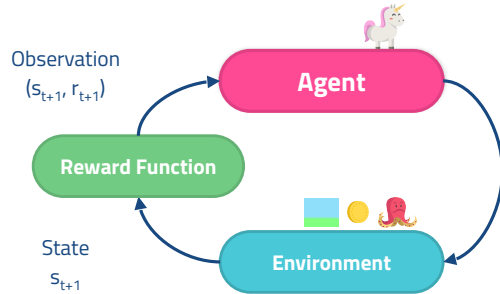
DEMO



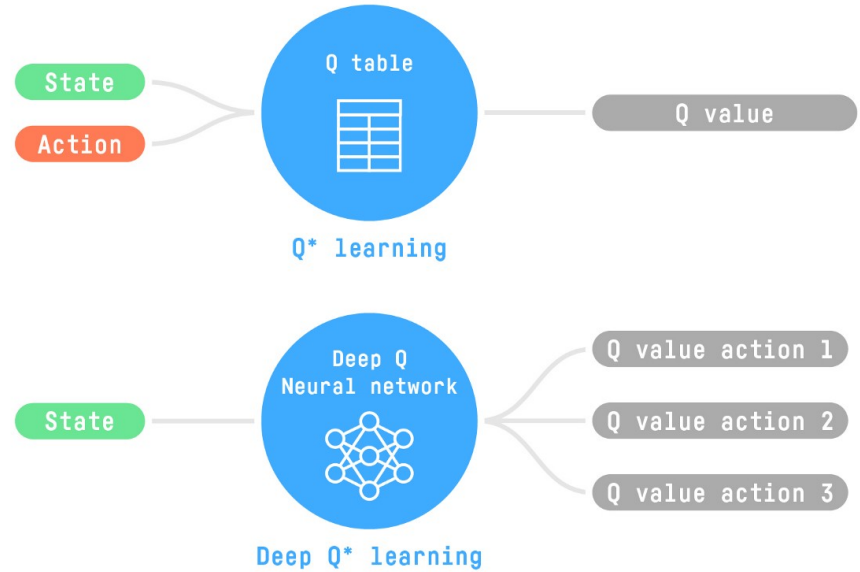
<https://github.com/IngyN/pyladies-talk>

Reinforcement Learning

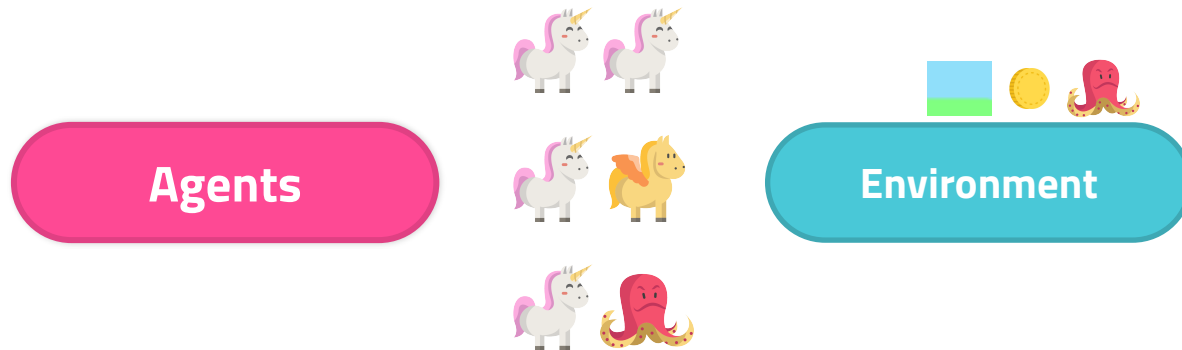
Deep Q-Learning



State, Action
 (s_t, a_t)



Multi-Agent Reinforcement Learning

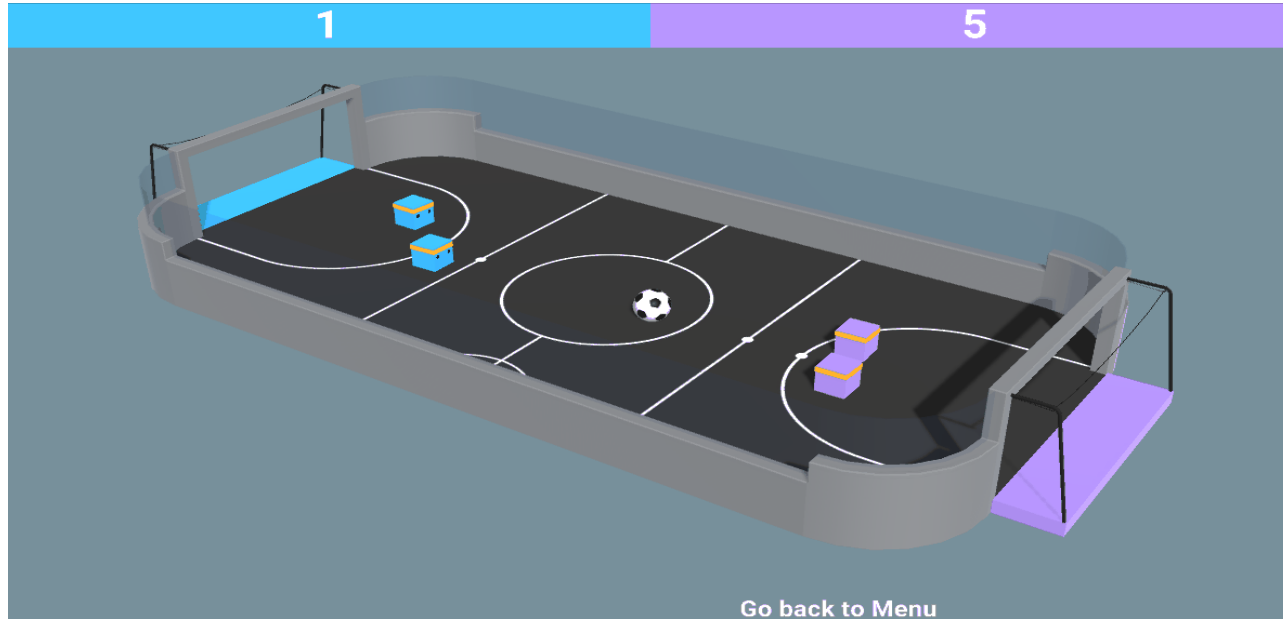


Coop / Versus

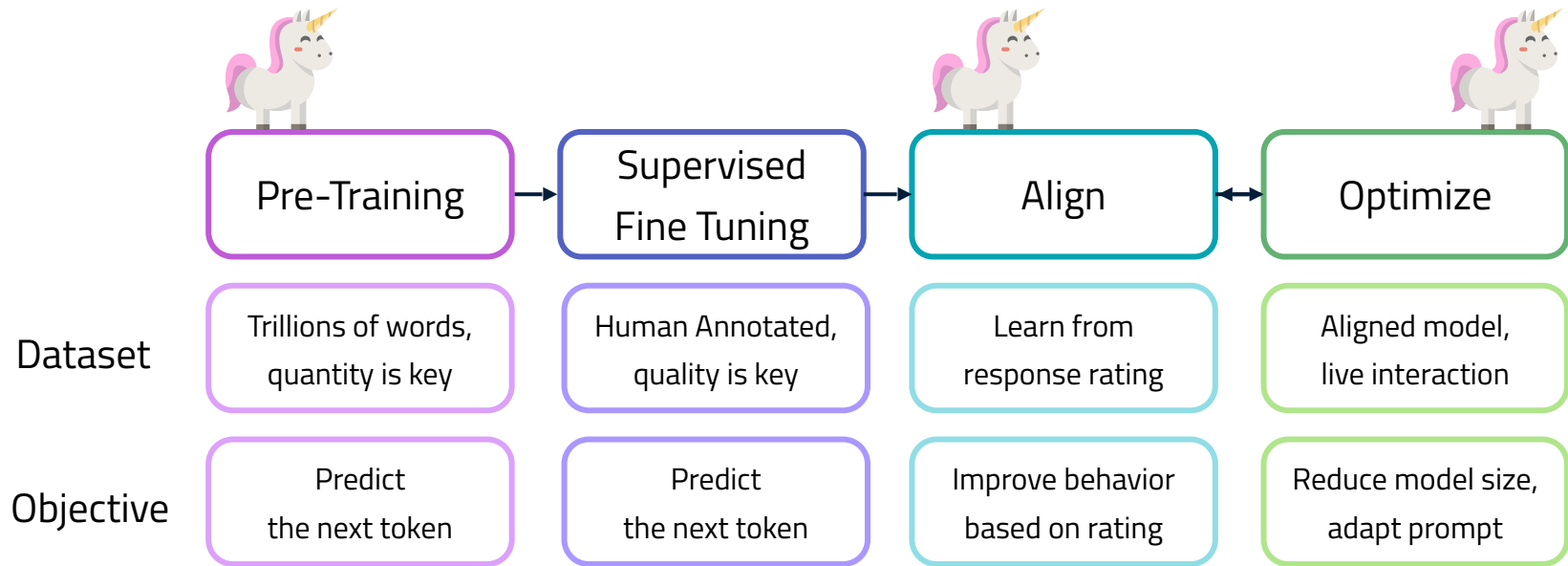
Credit Assignment

Scalability

Multi-Agent Reinforcement Learning



The LLM Training Pipeline

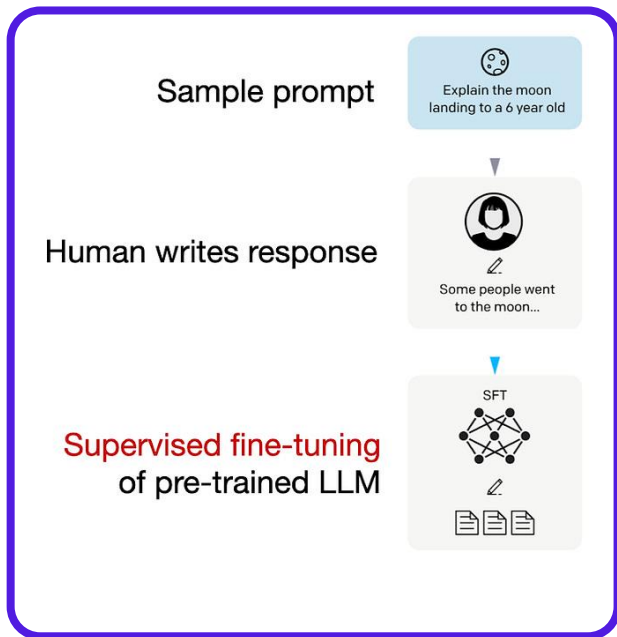




Align

Reinforcement Learning with Human Feedback (RLHF)

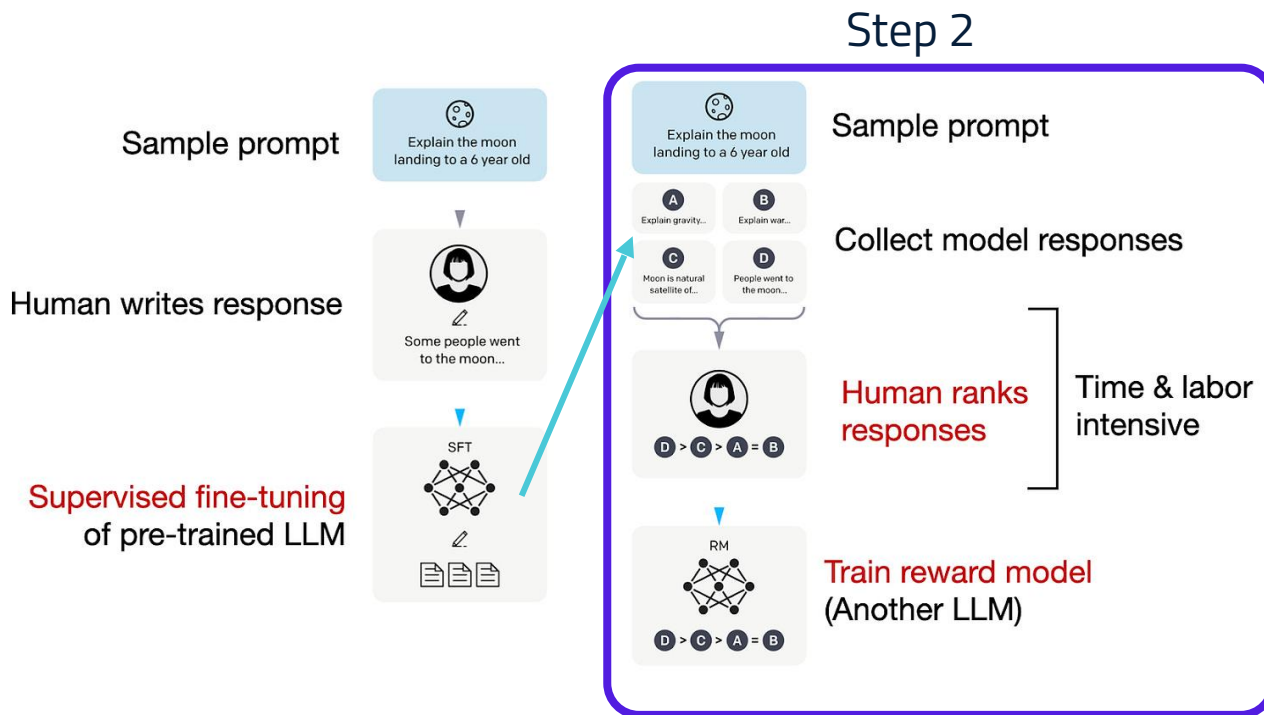
Step 1





Align

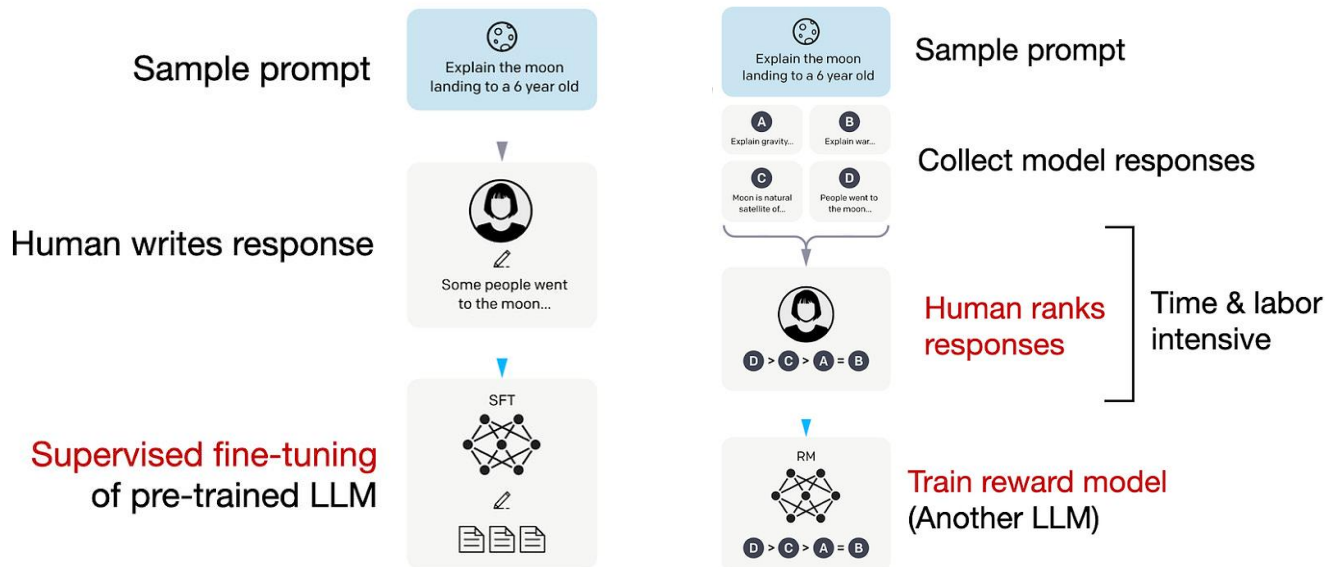
Reinforcement Learning with Human Feedback (RLHF)





Align

Reinforcement Learning with Human Feedback (RLHF)



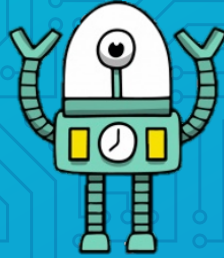
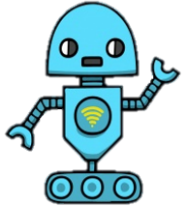
RL algorithm to update the LLM

Proximal Policy Optimization (PPO)

DEMO



<https://github.com/IngyN/pyladies-talk>



Thank you for your attention
Questions?



References

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Resources to learn more:

- [17] Ahead of AI Blog by Sebastian Raschka : <https://magazine.sebastianraschka.com/>
- [18] Hugging Face Course on RL : <https://huggingface.co/learn/deep-rl-course/en/unit0/introduction>
- [19] Berkeley RL Course : <https://inst.eecs.berkeley.edu/~cs188/archive/su25/>